Digital Image processing

Introduction

The basic aim of study of digital image processing is to process the image (in digital form) for—

1. improvement of image for human interpretation and
2. processing of image data for storage, presentation and transmission of image by machine.

1.1 DEFINITION OF IMAGE AND DIGITAL IMAGE PROCESSING

Before studying digital image processing, first of all we should be cleared about image.

"An image may be defined as a two dimensional function $f(x, y)$, where x and y are spatial (plane) co-ordinates. The amplitude of 'f' at any pair of co-ordinate (x, y) is called the intensity '*or*' gray level of the image at that point".

When (x, y) and amplitude values of 'f' are all finite, discrete quantities, we call the image a digital image.

Now we should define digital image processing. Digital image processing may be defined as a

1. "*Discipline in which both input and output of a process are image*".

   and

2. "*Process of extracting attributes from images (up to the level till we are able to recognize the individual object)*".

1.4 COMPONENTS OF AN IMAGE PROCESSING SYSTEM

Although the components of an image processing system differs according to different types of images, for example; satellite images and X-ray images cannot be processed by same type of image processing system. But to have a knowledge about the minimum requirement of an image processing system, we can study the block diagram given below of a general purpose image processing system.

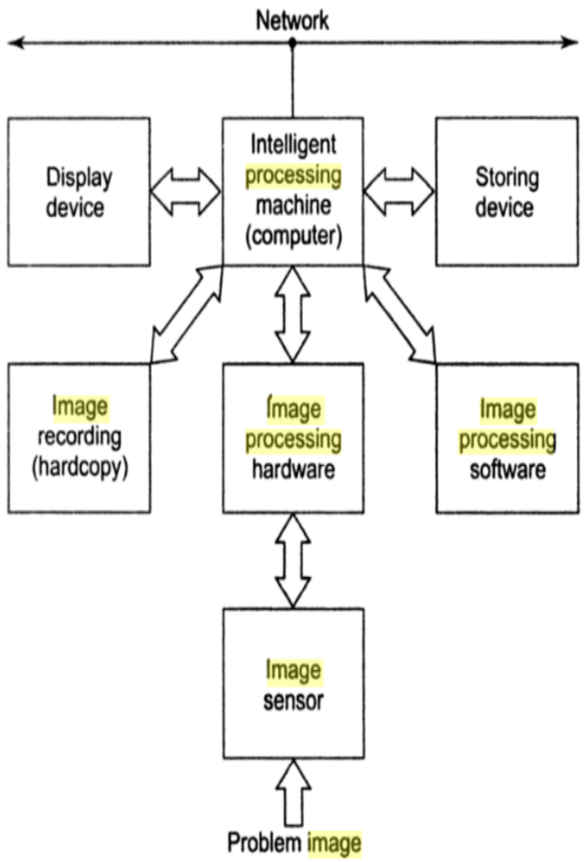


Fig. 1.3 *Components of digital image processing system.*

Now we study each part one by one :

1. **Image Sensor :** First basic requirement is image sensor. It has basically two parts :
   (*a*) A physical device that will sense the energy radiated by the object we wish to image.
   (*b*) A second device that is called digitizer that will convert the output of physical device into digital form.
2. **Image Processing Hardware :** Image processing hardware is here used just before the intelligent device (generally a super computer) because it will perform the specialized process on image but with high speed. For example, it may take average of different samples of an image for reduction of noise.
3. **Intelligent Processing Machine (Computer) :** It basically may be a general purpose PC *'or'* a supercomputer according to task to be performed. The basic use of this device is that it will perform all digital image processing task off line. Off line means once, we have taken image, we can apply different enhancement techniques (and other processes also).
4. **Image Processing Software :** Image processing softwares are specially designed programming modules that perform specific tasks. Some software also provides the facility that a user can also write programming codes for some purposes.
5. **Storing Device :** Storing devices are used for storing of an image for different purposes. On the basis of tasks, the storing device can be of three types :
   (*a*) Short-term storage for use during process.
   (*b*) On-line storage for relatively fast recall.
   (*c*) Storage for frequent use.
6. **Display Device :** Display device is generally a color monitor.
7. **Image Recording (Hard Copy) :** It may be only one of leser printers, inkjet units, CD-Roms etc.
8. **Networking :** By networking, one system user can also process the system at another place. For image, we require high bandwidth so optical fiber and broadband technologies are better options.

Fundamental Steps in DIP

1.3 FUNDAMENTAL STEPS IN DIGITAL IMAGE PROCESSING

Till now, it should be clear that the area of digital image processing is very broad. To study the digital image processing, the whole process has been divided into some fundamental steps as shown in Figure 1.2.

Output of these processes are generally images

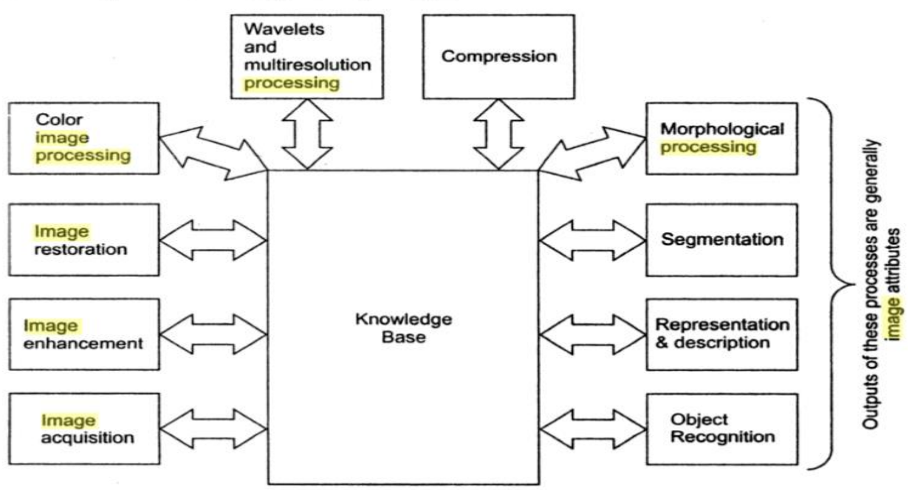


Fig. 1.2 *Fundamental steps in digital image processing.*

1. **Image Acquisition :** This is the first step of digital image processing. Definitely first step of digital image processing should be sensing of an image. So in an image acquisition, image is sensed by *'illumination'* from source and *'reflection'* or *'observation'* from sensors. It will also involves preprocessing, such as scaling. Example of image acquisition is to take an image by a camera.
2. **Image Enhancement :** After taking an image, it is some time required to highlight certain features of image. So image enhancement is simply to highlight certain features of image.

Example of image enhancement is simply to increase the contrast of image so that it looks better.

3. **Image Restoration :** Restoration means to restore an image that is not looking good due to some distortion (Noise). Thus, we can say that restoration also involves the process so that image looks better just like image enhancement. But image enhancement is a subjective process that is based on human perception only. Here we see that image appearance is increasing but we don't bother about *'why'* and *'how'*?

   On the other hand, image restoration is a objective process. It is based on mathematical and probabilistic models of image degradation. Here we analysis *'why'* and after getting the reason of image degradation, we remove that by developing some methods (after using some mathematical analysis).

4. **Color Image Processing :** Color image processing deals with modeling and pocessing of color images.
5. **Wavelets and Multiresolution Processing :** Wavelets are the foundation for representing images in various degree of resolution.
6. **Compression :** Compression is a technique that is used for reducing the storage space for saving an image and reducing the bandwidth required for transmission of image.
7. **Morphological Processing :** It basically deals with tools for extracting image components that are useful for representation and description of shape. Definitely the outputs of this process are image attributes.
8. **Segmentation :** Segmentation is the process in which image is converted into small segments so that we can extract the more accurate image attributes. If the segments are properly autonomous (two segments of an image should not have any identical information) then representation and description of image will be accurate and if we are taking rugged segmentation, the result will not be accurate.
9. **Representation and Description :** Representation involves the steps that extracts the attributes that are useful for processing through computer. It can be—
   (*a*) Boundary representation that causes shaping of corners and inflections. It also separates one image from another.
   (*b*) Regional representation that is basically focused on internal properties of image such as texture, contrast etc.

   Description is also called feature selection. As name implies it deals with extracting the attributes that results in some quantitive information of interest. It can also be used for differentiating one class of objects from other.
10. **Recognition :** Recognition is a process that assigns a label to an object based on its descriptors.
11. **Knowledge Base :** Knowledge base can be defined as software that may help to user proper image enhancement, restoration 'or' compression technique. It may help user in segmentation of an image.

1.5.1 Image Formation in Eyes

The principle difference between the lens of eye and ordinary lens is that eyes lens is flexible. The shape of the lens is controlled by ciliary body. To focus on distant objects, ciliary body muscles allows the lens to be relatively flattened. Similarly if object is very near, then these muscles allow the lens to be thicker.

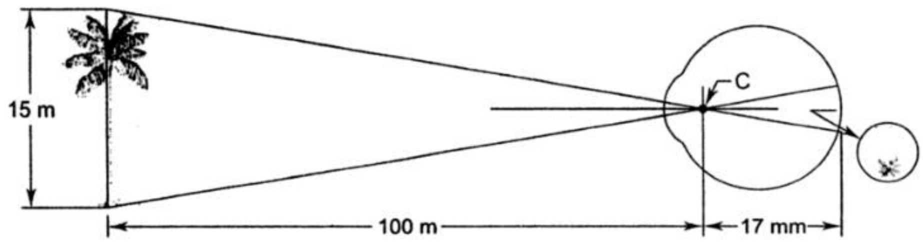


Fig. 1.6 *Image formation in eyes.*

The distance between the center of lens and the retina is called focal length. This length varies from 17 mm to 14 mm according to refractive power of lens. When object is away them about 3 mm, then lens has its lower refractive power. If the object is nearer then 3 mm, lens has its higher refractive index. By very small mathematical calculation, we can easily calculate the retinal height of image of object as for example :

$$\frac{15}{100} = \frac{h}{17}$$

Then $$h = 2.55 \text{ mm.}$$

1.6 IMAGE ACQUISITION

The images are generated by combination of an "*illumination*" source and the reflection 'or' absorption of energy from that source by the elements of the "*scene*" being imaged. Illumination may be originated by a radar, infrared energy source and X-ray energy source, ultrasound energy source, computer generated illumination pattern etc.

To sense the image, we use sensor according to the nature of illumination. The process of image sense is called image acquisition.

1.7 A SIMPLE IMAGE MODEL

Now we will proceed for mathematical representation of image. As we have already discussed, image can be presented by two-dimensional function $f(x, y)$ where (x, y) are the co-ordinates and amplitude of 'f' is called gray level '*or*' intensity level of image. Practically an image $f(x, y)$ must be non-zero and finite quantity ; that is

$$0 < f(x, y) < \infty$$

It is also discussed that for an image $f(x, y)$, we have two factors :

(*a*) The amount of source illumination incident on the scene being imaged. Let us represent it by $i(x, y)$.

(*b*) The amount of illumination reflected 'or' absorped by objects in the scene. Let us represent it by $r(x, y)$.

Then $f(x, y)$ can be represented by

$$f(x, y) = i(x, y) \cdot r(x, y)$$

Where $$0 < i(x, y) < \infty$$

it means illumination will be a non-zero and finite quantity and its quantity will depend upon illumination source

and $$0 < r(x, y) < 1$$

that indicate '*0*' means no reflection and total absorption and '*1*' means, no absorption and total reflection. The quantity of absorption '*or*' reflection depends upon characteristics of the imaged object.

1.7.1 Image Sampling and Quantization

We have acquired image by different sensor methods. But the response of all sensors is a continuous. Now to create a digital image, continuous sensed data should be converted into digital form. This involves two processes called sampling and quantization.

As we know the image is represented by $f(x, y)$, that is to be converted into digital form. An image $f(x, y)$ may be continuous with respect to x and y co-ordinates and also in amplitude. So,

1. "*Digitizing the coordinates values is called sampling*" ; and
2. "*Digitizing the amplitude values is called quantization*".

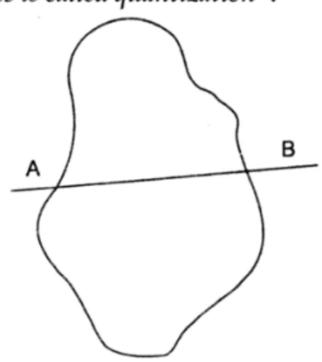


Fig. 1.12(a) *Test Image.*

Fig. 1.12(a) Test image.

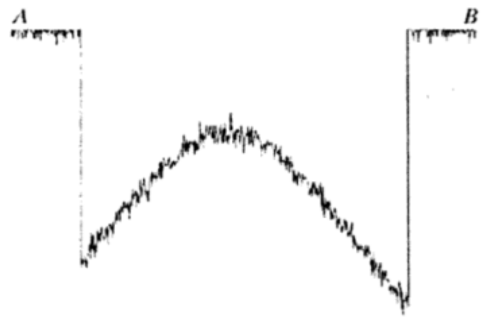


Fig. 1.12(b) *One-dimensional function of AB.*

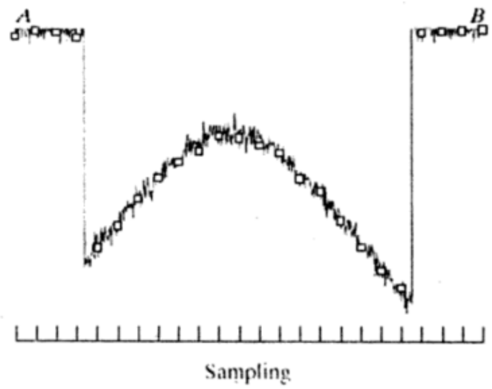


Fig. 1.12(c) *Sampled output.*

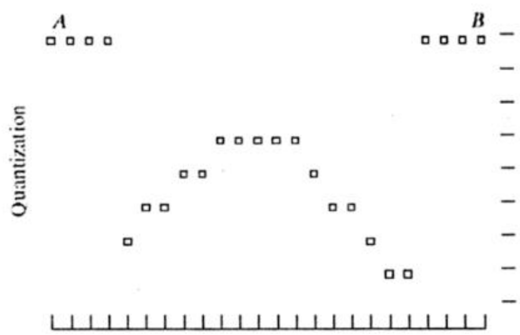


Fig. 1.12(d) *Quantized output.*

Here in Fig. 1.12 (*b*), we have drawn one dimensional function of line AB of an image shown in Fig. 1.12 (*a*). The random variations are due to noise. Now by sampling, we get the white squared output as shown in Fig. 1.12 (*c*). But still image is continuous on the gray scale. Finally after quantization we get complete digital output as shown in Fig. 1.12 (*d*).

If image is generated by single sensor ; the process of sampling and quantization will be same as shown in figure and discussed so far.

When image is generated by sensor strip ; the sampling is limited in one direction. But if image is taken by sensor array ; sampling is limited by both direction. Quantization process remains the same for all the methods.

1.7.2 Representation of Digital Image

The result of sampling and quantization is a matrix of real numbers.

Assuming that an image $f(x, y)$ is sampled so that resulting digital image has M rows and N columns. Now

(a) The co-ordinate at origin are

$$(x, y) = (0, 0)$$

(b) The co-ordinate at first row will be

$$(0, 0), (0, 1), (0, 2), \ldots\ldots (0, N-1)$$

and so on.

Here (0, 1) shows the second sample at first row. So we have matrix in the form given below :

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & \ldots\ldots & f(0, N-1) \\ f(1, 0) & f(1, 1) & \ldots\ldots & f(1, N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1, 0) & f(M-1, 1) & \ldots\ldots & f(M-1, N-1) \end{bmatrix}$$

Values of M and N are totally positive and cannot be negative.

Thus amplitude of 'f' called gray level has also been digitized in the range [0, L–1] means in total L levels. This range is called dynamic range of an image.

Here if we talk about the storage of an image in computer then we have to calculate the gray level in terms of bits. If number of bits are K, then gray levels will be

$$L = 2^K$$

It means if we say that an image has 256 gray level then it is an

$$256 = 2^K$$

or

$$K = 8$$

8 bit image.

1.8 ZOOMING AND SHRINKING OF DIGITAL IMAGE

Zooming may be viewed as oversampling, while shrinking may be viewed as undersampling. Definitely both can be applied tc the digital image only. In the zooming, we simply create new pixel location and assign gray levels for these new location.

If zooming is done by non-integer times (say 1.5 times to original image) ; then for assigning new gray level we follow nearest neighbour interpolation. It means for example we have 500×500 pixel image that is zoomed by 1.5 times. Due to this zooming we get new image of 750×750 pixel. Now new pixels are assigned gray level according to nearest gray level.

If zooming is done by integer times (say 2 times) ; then we may use pixel replication method. It means we duplicate each column if we are zooming image in horizontal direction and we duplicate each row, if we are zooming image in vertical direction.

For shrinking, we apply same process but in reverse direction (in place of duplication of pixel values, we delete rows and columns).

Summary

1. The aim of digital image processing is the improvement in the quality of image and make it suitable for different processes such as storage, transmission 'or' presentation.
2. Image is defined by two-dimensional function $f(x, y)$, where x, y are spatial (plane) co-ordinates and amplitude of $f(x, y)$ is called intensity level.
3. There may be following fundamental steps of image processing—Image Acquisition, Image Enhancement, Image Restoration, Color Image Processing, Wavelet and Multiresolution Processing, Image Compression, Morphological Processing, Image Segmentation, Representation and Description and Object Recognition.
4. The components of an image processing system may consist of the following parts :

   Image Sensor, Image Processing Hardware, Intelligent Processing Machine, Display device, Storing Device, Image Recording System and Image Processing Software etc.
5. Image acquisition means—'*illumination*' from energy source and '*reflection*' '*or*' absorption by object to be imaged.
6. Acquisition is done with the help of image sensor. Sensor may be a single image sensor, strip type sensor '*or*' array type sensor.
7. After taking image by a sensor ; image is digitalized with the help of digitalizer.
8. Digitalizing of co-ordinate values is called '*Sampling*'.
9. Digitalizing of amplitude values is called '*Quantization*'.
10. Zooming may be viewed as '*Oversampling*' and shrinking may be viewed as '*undersampling*'.

Image Enhancement in spatial domain

Image enhancement's approaches falls two categories :

(*a*) **Spatial Domain :** Spatial means the plane of image itself. So in spatial domain is based on direct manipulation of pixels of an image. Term spatial domain refers to the aggregate of pixel composing an image.

(*b*) **Frequency Domain :** Frequency domain processing techniques are based on manipulation in Fourier transform of image.

Enhancement is a subjective matter. The image can be '*good*' for one human being and can be '*bad*' for other person. Thus, for applying any enhancement technique, first of all we apply trial and error method.

2.2 MATHEMATICAL ANALYSIS OF ENHANCEMENT IN SPATIAL DOMAIN

1. Spatial domain analysis is applied directly to the pixels of image so it can be shown by

$$\boxed{g(x, y) = T[f(x, y)]} \qquad \text{... (2.1)}$$

$f(x, y)$ = Input image

$g(x, y)$ = Output image.

T = Operator of $f(x, y)$ whose value depends the neighbourhoods of (x, y).

2. **Gray-level transformation (Intensity Mapping) :** It is the simplest form of spatial domain. If we take the value of '*T*' when the neighbourhoods is size of 1×1 (single pixel). In this case, $g(x, y)$ depends upon $f(x, y)$ only. In this case, this is called gray-level transformation '*or*' mapping. It is represented by:

$$S = T(r) \qquad \text{... (2.2)}$$
$$S = g(x, y)$$
$$r = f(x, y)$$

3. **Contrast Stretching :** The spatial domain equation (2.2) shows basically the image we get after equation (2.2) would be of high contrast.

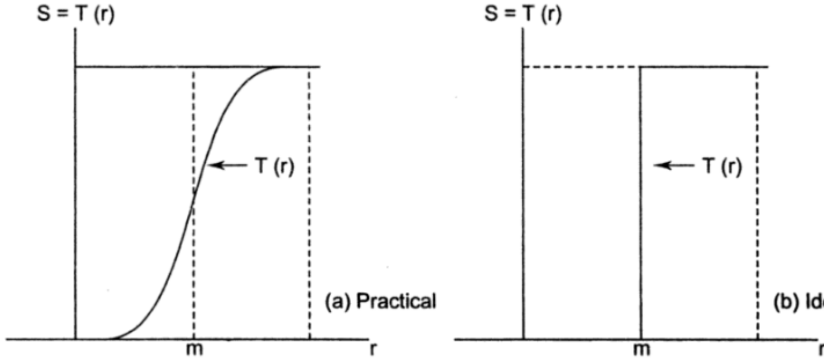


Fig. 2.1 *Contrast stretching and thresholding.*

4. **Thresholding :** The result of equation 2.2 is also thresholding of image. If pixel has value less then '*m*', we get dark pixel and if pixel has value high then '*m*', we get light pixel.

Because enhancement at any point in an image depends only on the gray level at that point, so this technique is also referred as point processing.

2.3 BASIC GRAY LEVEL TRANSFORMATION

Basically we have different types of gray level transformations

1. Point transformation.
2. Linear (negative and identity) transformation.
3. Logarithmic (log and inverse log) transformation.
4. Power-low transformation.
5. Piece-wise linear transformation.

2.3.1 Point Transformation

By equation,

$$S = T(r) \qquad \text{(By equation 2.2)}$$

where T = transform operator, this can be done by using 1×1 window that means only a single pixel; that type of transformation is called point transformation.

Basically use of point transformation is masking of a particular pixel.

2.3.2 Linear Transformation

Positive *'or'* identity transformation does not play an important role in image processing, because transformed image is image itself.

Negative transformation plays a vital role in image processing. Suppose we have total number of gray level = L

Then the range of gray levels = [0, L – 1] so the negative imaging can be presented by equation,

$$\boxed{S = (L - 1) - r} \quad \text{... (2.3)}$$

The application of negative imaging is to create a negative of image. If black area is dominant area in image that can be converted into white area and very small but important white in real picture is now converted into black spot. So now,

Black area with small spot white = White area with small spot black

Underlined image can be analysed easily.

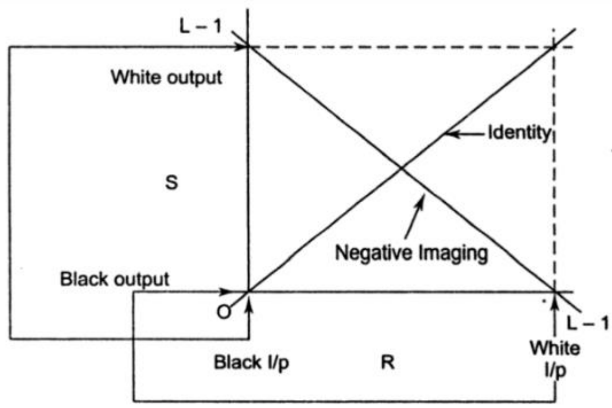


Fig. 2.2 *Line or transformation.*

2.3.3 Log Transformation

A general equation for log transformation is given by

$$\boxed{S = c \log(1 + r)} \quad \text{... (2.4)}$$

where c = constant and

$r \geq 0$.

The transfer characteristics has been drawn below :

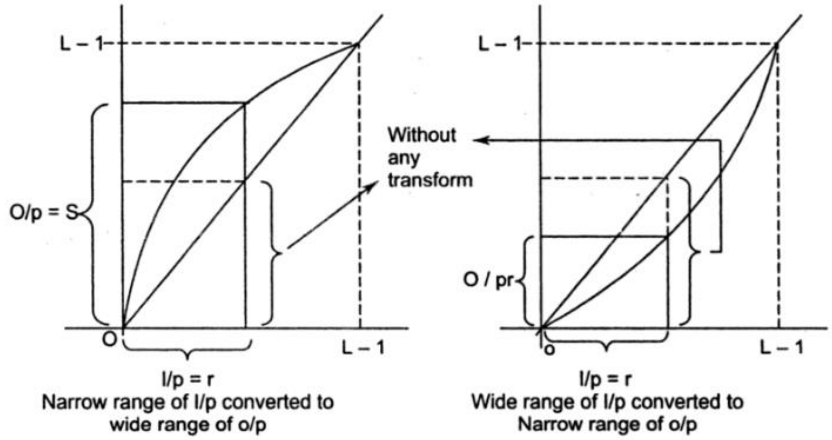


Fig. 2.3 *Log transformation.*

That shows that by log transformation, we get spreading / compressing of gray levels (that can also be done by power low transformation). But the actual advantage of log-transformation is compression and expansion of an image.

2.3.4 Power Low Transformation (nth power and nth root transformation).

The basic equation for power low transformation is

$$S = cr^{\gamma} \qquad \text{... (2.5)}$$

transfer function can be drawn as given below :

(*i*) for $\gamma > 1$ It is n^{th} power

(*ii*) for $\gamma < 1$ It is n^{th} root of pixel.

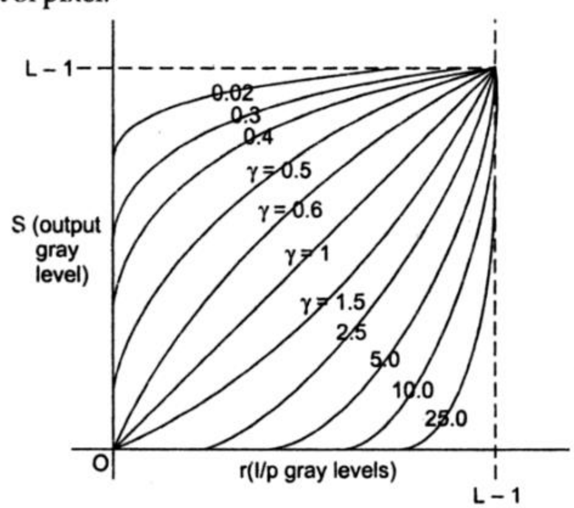


Fig. 2.4 *Power low transformation.*

(also called gamma correction due to the use of gamma as constant). Basically γ correction plays same role as log-transformation ; but now with small variation in the value of γ causes a long range (dynamic range) coverage. The most popular advantage of gamma level-correction is in CRT.

As at internet, a single picture in a website is visited by different peoples having different CRT. So many CRT has automatic gamma correction and some have manual that carries to see the image according to our choice and our monitor.

2.3.5 Piece-wise Linear Transformation

In the transformations discussed till now has one disadvantage that "*transformation is applied to whole image. On the other hand, if we require to enhance a particular part of image, we generally prefer a piece-wise linear transformation*".

Now we will discuss different types of piece-wise transformation functions and their applications.

(*a*) **Contrast Stretching :** Basically due to different reasons such as :

(*i*) Poor illumination.

(*ii*) Lack of dynamic range in imaging sensor.

(*iii*) Wrong setting of lens aperture during image acquisition.

We get poor contrast, so to get good contrast, we apply contrast stretching. The transfer function has been drawn below :

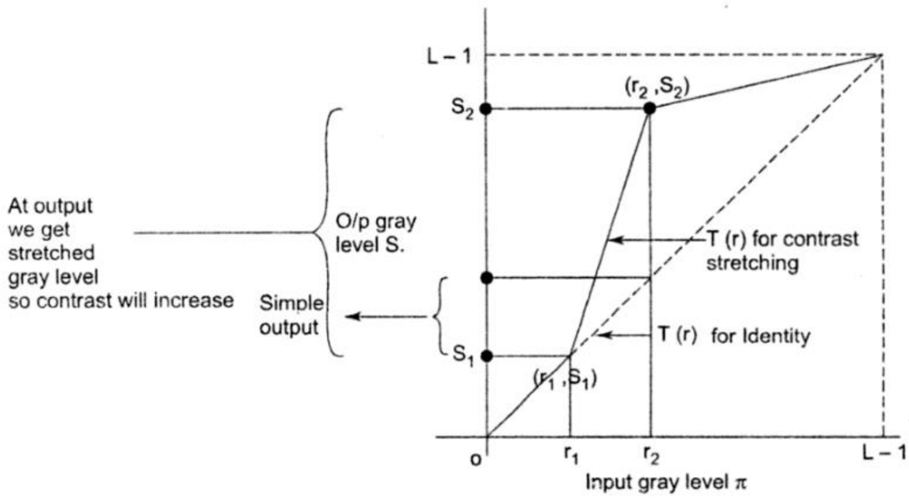


Fig. 2.5 *Contrast stretching.*

This figure clearly shows that a range of input gray levels $(r_1 - r_2)$ has been stretched $(s_1 - s_2)$ by piece-wise linear transform. If we don't apply piece wise we get normal unstretched output image.

If $\qquad (r_1, s_1) = (r_{min}, 0)$

and $\qquad (r_2, s_2) = (r_{max}, L - 1) \qquad$ ... (2.6)

We will simply threshold mapping (that gives us digital binary image).

(b) **Gray Level Slicing :** Sometime, it is desirable to stretch the specific range of image should be enhanced not the whole image.

So gray level slicing provide this facility, by two possible ways.:

(i) One approach is that we boast up the desired range of gray levels and other gray levels are suppressed.

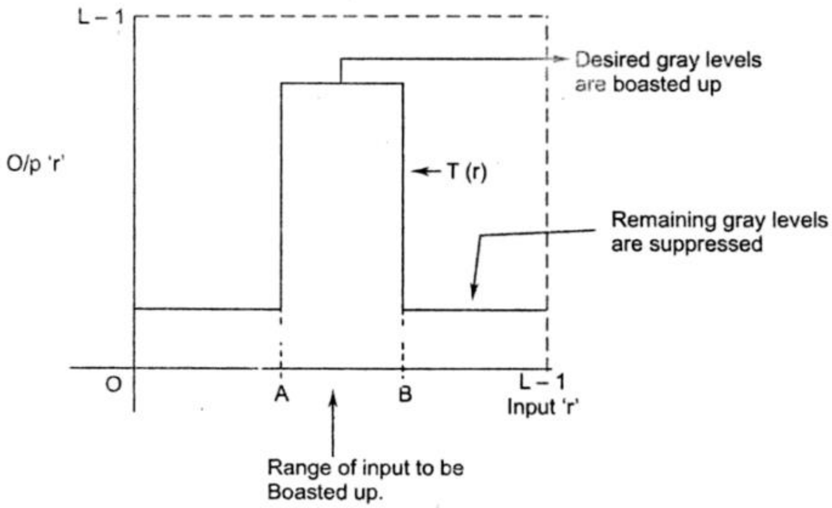


Fig. 2.6(a) *Gray level slicing.*

(ii) Another approach is that, we boast up the desired range of input, but also preserve the quantity of remaining port of input.

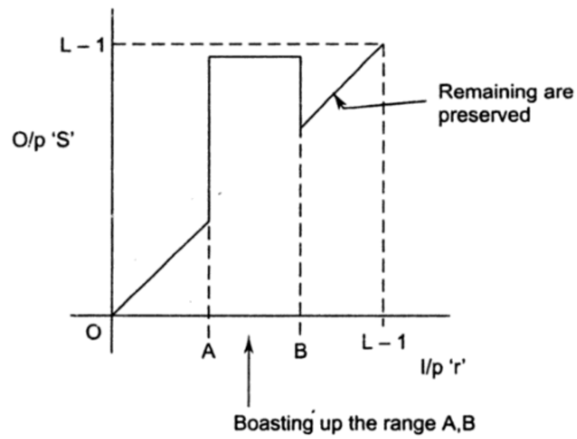


Fig. 2.6(b) *Gray level slicing.*

(*c*) **Bit-Plane Slicing :** Sometime it is also desired to know the contribution of each bit in an image. Suppose that we have 8 bits in each pixel.

Pixel having 8 Bits
$$\begin{bmatrix} a_{11} & a_{12} & \cdots\cdots & a_{1N} \\ a_{21} & a_{22} & \cdots\cdots & a_{2N} \\ \vdots & \vdots & & \vdots \\ a_{M1} & a_{2M} & \cdots\cdots & a_{MN} \end{bmatrix}_{M \times N} \qquad \ldots (2.7)$$

So it can be divided into 8-plane. Plane 0 will have all LSBs and plane '7' will have all MSB.

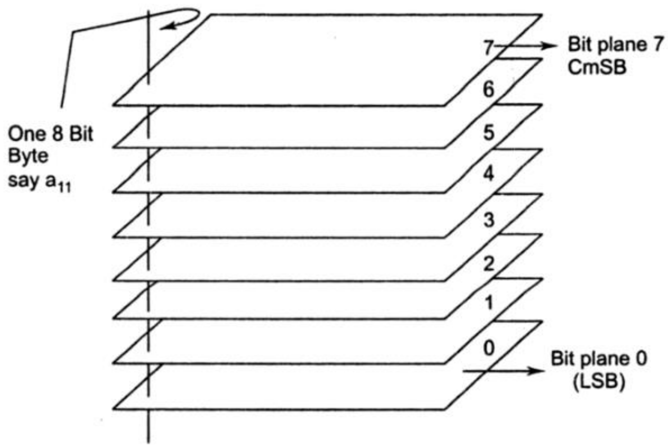


Fig. 2.7 *Bit plane slicing.*

It should be cleared that all LSB means plane '0' will play very less significant role and plane '7' will play very much significant role in image.

We may follow the following steps for bit slicing:

1. We have total $2^8 = 256$ levels. So by a appropriate threshold divide the levels 0 – 127 and 129–255.
2. Thus, we get two segments, again make threshold and subdivide the levels.

2.5 HISTOGRAM EQUALIZATION

"In histogram equalization, we get the image that has equalized gray levels".

By the equation,

$$S = t(r) \quad 0 \le r \le 1 \text{ for} \qquad \text{(by equations 2.2)}$$

binary scheme for histogram equalization transformation function must satisfy the following conditions.

(*a*) T (r) should be single valued that sures that the inverse transform will exist. and T (r) should be monotonically increasing that assure us that image is moving from black to white.

(*b*) and $0 \le T(r) \le 1$ that for $0 \le r \le 1$ so that input and output image have same size.

Now our basic aim is to get the

1. Input signal histogram.
2. A transfer function that will generate a equalized histogram of input signal.
3. To verify whether the output histogram generated is equalized '*or*' not.

Input signal histogram can be generated using probability density function (PDF) because the gray levels in a image are random distributed and PDF is only the tool by which we can calculate the histogram. It can be shown that if

$P_r(r)$ = PDF of input image gray levels.

and $T(r)$ = The transfer function for equalization are known, we have output $P_s(s)$ by relation,

$$P_s(s) = P_r(r) \left| \frac{dr}{ds} \right| \qquad \text{... (2.10)}$$

As we are using PDF for generating the histogram of input signal, the transform function should be in form of CDF (commulative distribution function).

So we have a new relation

$$S = T(r) = \int_0^r P_r(w)\, dw \qquad \text{... (2.11)}$$

w = dummy variable for integration

Now to verify that this T (r) will generate a output image that has equalized histogram ; let us calculate

$$\frac{ds}{dr} = \frac{d}{dr}\left[\int_0^r P_r(w) \, . \, dw\right]$$

$$\frac{ds}{dr} = P_r(r)$$

or $$\left|\frac{dr}{ds}\right| = \frac{1}{P_r(r)} \quad \text{... (2.12)}$$

Substituting equation (2.12) in equation (2.11), we have

$$P_s(s) = P_r(r) \times \frac{1}{P_r(r)}$$

$$P_s(s) = 1 \quad \text{... (2.13)}$$

That means output is totally independent upon input. It will generate a equalized histogram of an image.

Thus, if we apply any image to equation (2.11) we will get equalized histogram of image.

2.6 HISTOGRAM MATCHING

In histogram equalization, we get the output image that has equalized gray levels. But sometime, it is required that we want a specific shape of histogram of image.

So *"to generate a image that has the histogram of our desire is known as histogram matching 'or' histogram specifications"*.

Mathematical Analysis

Suppose that we have image 's' and PDF of it is $P_s(s)$. Now we want to generate a image 'z' with PDF $P_z(z)$ of specific value. So, we know that

$$S = T(r) = \int_0^r P_r(w)\, dw \quad \text{... (2.14)}$$

Again for specific

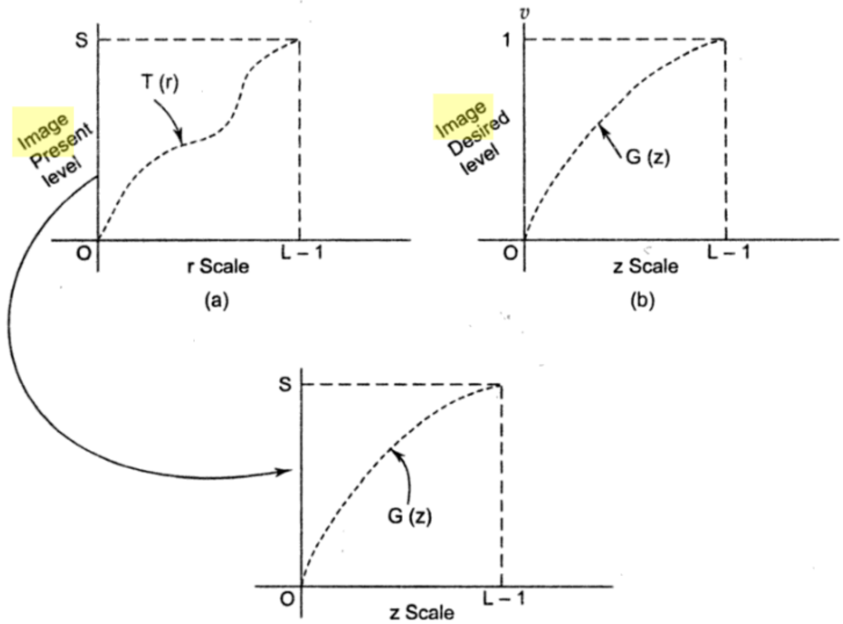
$$S = G(z) = \int_0^z P_z(k)\, dk \quad \text{... (2.15)}$$

'S' is finally the basic image for both.

So, $$Z = G^{-1}(s) = G^{-1}[T(r)] \quad \text{... (2.16)}$$

is the required image. For this we should follow the slips:

1. Calculate $T(r)$ for original image.
2. Calculate $G(z)$ for desired image.
3. Now apply equation (2.16) to get that desired values of 'z'. Now for each value of z calculate s by equation (2.15). Graphically, the method is shown on next page :



(c) Now - superimpose the s & w scale for G(z) & calculate new z

Fig. 2.9 *Process of histogram matching.*